

The Rope Winding Angle as the Weinberg Angle

A Geometric Estimate $\sin^2\theta_W = 1/(3\sqrt{2})$, and Its ~2% Gap

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Naming note (5 September 2026). This programme is now the Mesh Programme and the theory it develops the mesh framework; 'the rope framework' in earlier revisions reads 'the mesh framework'. 'Rope' remains the name of the two-strand wound object, and the Rope Hypothesis credited to Bill Gaede keeps its name. Identifiers (repository, rope_solver, file names, claim IDs) are unchanged.

Abstract

The mesh framework proposes that the electroweak mixing (Weinberg) angle is set by the winding geometry of the two-strand rope, giving $\sin^2\theta_W = 1/(3\sqrt{2}) = 0.2357$ — the '3' from the three spatial dimensions and the ' $\sqrt{2}$ ' from the two strands. This estimate sits about 1.94% from the measured value 0.23122. We report it as a geometric conjecture of the correct magnitude and structure, and we are explicit that the ~2% gap is real and consequential: it is the input that, when used in the lepton-mass relation, breaks it. All numbers are outputs of the rope_solver / tests computation in this package, not inserted constants.

Scope of this paper

CLAIM: a geometric estimate $\sin^2\theta_W = 1/(3\sqrt{2}) = 0.2357$ from two-strand rope winding, ~2% from measurement.

NOT CLAIMED: a precision derivation. The ~1.94% gap is real. Status: DEMOTED 5 September 2026 (KOIDE-WEINBERG, EW-001) to a KEPT COINCIDENCE with three named choice points; see the Disposition section.

1. Winding Geometry and the Mixing Angle

The Weinberg angle parametrises the mixing of the electroweak gauge fields. In the mesh framework it is read from the winding geometry of the two-strand rope: three spatial dimensions supply the factor 3, the two strands supply the $\sqrt{2}$, giving $\sin^2\theta_W = 1/(3\sqrt{2})$.

$$\sin^2\theta_W = 1/(3\sqrt{2}) = 0.2357 \quad (1.1) \quad \text{— measured: } 0.23122$$

2. The Gap, Stated Plainly

The honest status of this value. The predicted $\sin^2\theta_W = 1/(3\sqrt{2}) = 0.2357$ differs from the measured 0.23122 by about 1.94%. It is close, and its form is suggestive, but it is NOT exact, and the test suite pins it explicitly as a 'soft external input' — the value that, when fed into the lepton-

mass relation instead of the measured angle, breaks it. This is therefore a geometric conjecture of the right magnitude, not a precision derivation, and the ~2% gap is the known weak link of the electroweak sector.

Disposition (5 September 2026): demoted to a kept coincidence

Verdict. Commission KOIDE-WEINBERG (ITEM 6d) executed the derivation-or-demotion this paper's status line deferred. Under FND-088's protocol (require a dependency path from registered objects to the number, with every choice point listed, before looking at the target) the chain 'three dimensions supply the 3, two strands supply the $\sqrt{2}$ ' has three choice points on its face: (1) the combination rule, why $1/(3\sqrt{2})$ rather than $1/\sqrt{6}$, $1/3^2$, $\sqrt{2}/3$ or $1/(3+\sqrt{2})$, which the two-axioms paper itself records as 'uniqueness is Open'; (2) the quantity, why $\sin^2\theta_W$ carries the geometric number rather than $\sin\theta_W$, $\tan^2\theta_W$ or g'/g ; (3) the multiplicative reading of the two factors, for which the Hopf route supplies transparency but no forcing mechanism. A chain with choice points is a conjecture whose choices are now named. The registered verdict is EW-DEMOTE: EW-001 leaves Conjecture grade and is kept as a coincidence. Full record: analysis/KOIDE_WEINBERG_results.md.

The renormalization point, fixed before re-reading. A scale-free geometric number can be compared to exactly one renormalization point. The registered comparison, 0.23122 (the MS-bar value at M_Z), was locked before the comparison was re-read; against it the estimate is 1.94% off. The low- Q^2 effective angle (about 0.238) was displayed alongside and never adjudicated against: the estimate is 0.97% from it. A number within one to two percent of whichever point one picks is what a coincidence at that level looks like, and running or threshold effects are not invoked to close the gap.

What stands. The two structural readings (three dimensions, two strands) stand as readings; the 1.94% gap stands as measured; the test-suite pin stands. What is withdrawn is the word 'estimate' as a claim about the electroweak sector: the corpus holds no derived Weinberg angle. The companion lepton-mass relation (PM-001) was demoted in the same commission on independent grounds (its sensitivity to this angle is 71: a 1.94% change in the input moves the muon-to-electron mass ratio by a factor 7.8), which closes the coupled inconsistency the corpus carried between the two entries.

Status

$\sin^2\theta_W = 1/(3\sqrt{2}) = 0.2357$ geometric estimate: DEMOTED to a kept coincidence (EW-001, 5 September 2026); three choice points, no derivation.
~1.94% deviation from measured 0.23122 (0.97% from the low- Q^2 value, displayed only); the companion Koide relation was demoted with it.

Registered Claims in This Paper

Each statement below is traceable to the machine-readable registry (claims.yaml), the corpus's single source of truth. Status labels: Derived (follows from stated mechanics), Modeled (reproduces data with declared inputs), EFT-constrained (bounded by effective-theory limits), Conjecture (proposed, not derived), Open (registered unsolved problem), Failed (falsified, kept as a finding). Where a claim is code-backed, the named benchmark reruns it.

- **EW-001 [Failed: demoted 5 Sep 2026, kept coincidence]:** Weinberg angle $\sin^2\theta_W = 1/$

Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.